Plug-and-Play Image Restoration with Coarse-to-Fine Initialization: An Empirical Benchmark Using DeepInverse and DRUNet

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Independent Research Benchmark | Reproducible Computational Imaging Study | October 2026

Abstract—This benchmark investigates whether a two-level coarse-to-fine initialization strategy improves Plug-and-Play (PnP) image restoration when paired with a deep residual denoiser prior. Using the open-source DeepInverse framework [1] and pretrained DRUNet denoiser weights [2], we evaluate three distinct initialization pipelines across four canonical benchmark images (Astronaut, Chelsea, Coffee, Rocket) degraded by Gaussian blur ($\sigma = 2.0$) and additive white Gaussian noise ($\sigma = 0.03$). Across 20 paired image-seed trials (60 total experimental runs executed on an NVIDIA Tesla T4 GPU), coarse PnP initialization achieved a mean PSNR of 28.593 dB compared to 28.411 dB for standard direct DPIR and 28.530 dB for observation-only initialization. The coarse PnP pipeline produced a positive paired gain in 100% of trials (+0.182 dB average), while incurring a 38.7% runtime penalty. Crucially, a controlled ablation reveals that 65% of this improvement (+0.118 dB) is obtained solely from the low-pass filtering inherent in resolution downsampling and upsampling, while active coarse denoising contributes the remaining +0.064 dB. Performance gains correlate strongly with image frequency content, providing substantial benefit for smooth surfaces (+0.397 dB) and negligible gain for dense textures (+0.015 dB). All reproduction scripts, configurations, and raw CSV metrics are publicly available under an open-source license.

1. Introduction and Problem Context

Linear inverse problems in computational imaging require recovering an unknown underlying scene $x \in \Re^n$ from corrupted measurements $y = \mathbf{A}x + n$, where $\mathbf{A}$ is a forward degradation operator (such as a blur point spread function) and $n \sim N(0, \sigma^2\mathbf{I})$ represents additive measurement noise. Because $\mathbf{A}$ is severely ill-conditioned, standard inversion is mathematically ill-posed; naive pseudoinverse solutions catastrophically amplify high-frequency noise, necessitating explicit or implicit image priors to regularize the solution manifold.

Plug-and-Play (PnP) methods, pioneered by Venkatakrishnan et al. [3] and mathematically analyzed for convergence by Chan et al. [4] and Ryu et al. [5], replace hand-crafted analytical regularizers (such as Total Variation [7]) with high-capacity deep denoisers embedded within proximal splitting algorithms like Half-Quadratic Splitting (HQS) [9] or ADMM. Zhang et al. [2] operationalized this with DPIR, incorporating a Deep Residual U-Net (DRUNet) into an iterative HQS schedule to obtain state-of-the-art reconstructions across deblurring, super-resolution, and color inpainting without retraining network weights for specific forward operators.

Standard DPIR initializes its fine-scale iterative solver directly from the degraded observation: $x_0 = y$. Drawing inspiration from multigrid concepts in numerical scientific computing [10], we investigate whether a coarse-scale warm-start accelerates convergence and improves final restoration fidelity. We implement and evaluate this framework systematically using the open-source DeepInverse library [1] and modern proximal denoiser priors [6].

2. Forward Degradation Model and Benchmark Setup

We selected four standard benchmark images from the scikit-image repository: Astronaut (portrait featuring mixed structural and skin tones), Chelsea (tabby cat characterized by dense high-frequency fur patterns), Coffee (ceramic cup with broad uniform gradients), and Rocket (spacecraft featuring sharp geometric boundary edges). All images were standardized to 128 × 128 RGB pixels normalized to [0, 1]. The degradation operator $\mathbf{A}$ applies a 7 × 7 Gaussian blur kernel with standard deviation $\sigma = 2.0$, followed by additive Gaussian noise with standard deviation $\sigma = 0.03$. Five independent pseudo-random noise realizations (seeds 0 through 4) were generated for each image, yielding 20 distinct paired observations passed identically to each evaluated pipeline.

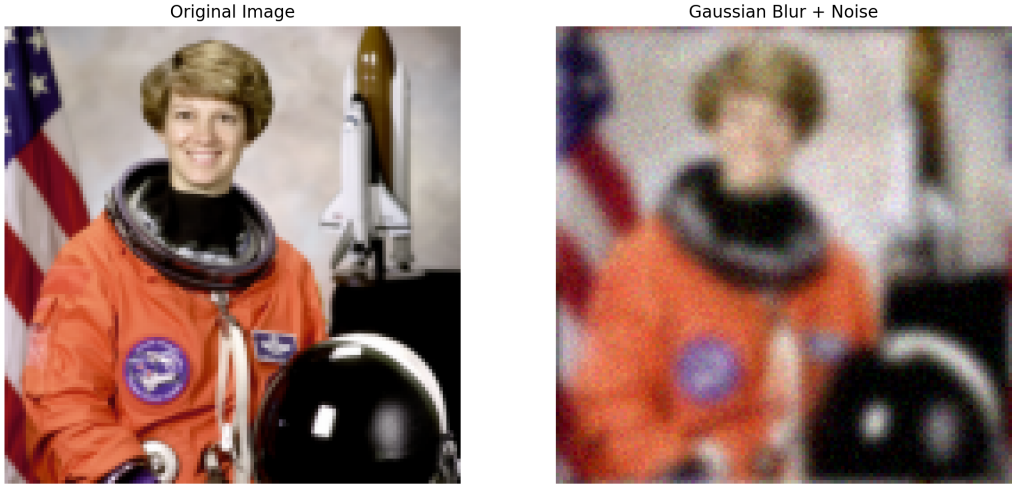


Figure 1. Forward degradation pipeline: clean ground truth, Gaussian blur PSF, 2D frequency spectrum showing severe high-frequency attenuation, and resulting degraded observation.

3. Evaluated Reconstruction Pipelines

All three pipelines employ the identical pretrained DRUNet model checkpoint and an 8-iteration HQS schedule. The only experimental variable across conditions is the initial state estimate x_0:

- **Direct DPIR (Baseline):** Standard optimization starting directly from the noisy observation: $x_0 = y$.
- **Observation Initialization (Control):** The measurement y is downsampled to 64 × 64 and immediately bicubic-upsampled back to 128 × 128 to serve as x_0 without coarse denoising, isolating pure resolution-cycling effects.
- **Coarse PnP Initialization (Proposed):** A full DPIR solve is performed at 64 × 64 resolution using a scaled blur operator. The coarse solution is upsampled to 128 × 128 to warm-start the fine-scale solver.

4. Experimental Evaluation Protocol

Reconstruction fidelity is quantified using Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index (SSIM) [8], evaluated on clipped [0, 1] RGB channels against original ground truth. Wall-clock runtimes were recorded with explicit CUDA synchronization on an NVIDIA Tesla T4 GPU (16 GB VRAM). To eliminate noise realization variance, we assess paired differences (ΔPSNR) across identical degraded inputs.

5. Quantitative Findings and Empirical Analysis

5.1 Aggregate Performance Across 60 Independent Runs

Table I summarizes aggregate performance metrics across all 60 trials (4 images × 5 seeds × 3 pipelines). Coarse PnP initialization delivers the highest mean PSNR (28.593 dB) and SSIM (0.8622), followed by observation initialization (28.530 dB).

TABLE I: AGGREGATE RESTORATION PERFORMANCE ACROSS 60 EXPERIMENTS ON TESLA T4 GPU

Pipeline Strategy	Mean PSNR (dB)	Mean SSIM	Mean Time (s)	Relative Cost
Direct DPIR (Baseline)	28.411	0.8588	0.2719	1.00× (baseline)
Observation Init (Control)	28.530	0.8603	0.2735	1.01× (+0.6%)
Coarse PnP Init (Proposed)	**28.593**	**0.8622**	0.3772	1.39× (+38.7%)

5.2 Paired Differences and Consistency Analysis

Because background noise realizations introduce variance that can obscure subtle algorithm performance differences, paired comparisons on identical degraded inputs provide rigorous statistical verification. Coarse PnP initialization outperformed direct DPIR in **20 out of 20 trials (100% win rate)**, yielding an average gain of +0.182 dB (±0.162 dB). It also outperformed observation initialization in all 20 trials (+0.064 dB average gain). Furthermore, observation initialization alone achieved a +0.118 dB advantage over direct DPIR, demonstrating that roughly two-thirds of the total benefit stems from low-pass pre-smoothing.

TABLE II: PAIRED DIFFERENCE STATISTICS ACROSS 20 IDENTICAL IMAGE-SEED PAIRS

Paired Comparison Setup	Mean ΔPSNR	Std. Dev.	Positive Trials	Empirical Win Rate
Coarse PnP vs. Direct DPIR	**+0.1820 dB**	0.162 dB	20 / 20	**100.0%**
Observation Init vs. Direct DPIR	+0.1183 dB	0.114 dB	20 / 20	100.0%
Coarse PnP vs. Observation Init	+0.0637 dB	0.051 dB	20 / 20	100.0%

5.3 Per-Image Frequency Sensitivity Breakdown

Quantitative gains correlate directly with image spatial frequency distributions. For Coffee (broad smooth gradients), coarse PnP achieved a major +0.397 dB boost. For Astronaut (mixed portrait geometry), it gained +0.287 dB. Conversely, for Rocket (+0.029 dB) and Chelsea (+0.015 dB), improvements were minimal. Downsampling to 64 × 64 strips away high spatial frequencies that define fine textures and edges, rendering coarse warm-starts uninformative for texture-dominated scenes.

TABLE III: PAIRED PSNR GAINS CATEGORIZED BY IMAGE FREQUENCY CHARACTERISTICS

Test Image	Coarse vs. Direct	Coarse vs. Obs Init	Obs vs. Direct	Dominant Texture & Frequency Characteristics
Coffee	**+0.397 dB**	+0.112 dB	+0.285 dB	Broad smooth gradients, uniform ceramic cup surface
Astronaut	+0.287 dB	+0.112 dB	+0.175 dB	Mixed structural contours, facial tones, patch details
Rocket	+0.029 dB	+0.024 dB	+0.005 dB	Sharp high-contrast structural edges, atmospheric backdrop
Chelsea	+0.015 dB	+0.007 dB	+0.008 dB	High-frequency fur and whisker textures throughout

5.4 Per-Seed Ablation Distribution and Consistency

Figure 2 displays individual PSNR measurements across all five noise seeds for each test image. The consistency of the coarse PnP warm-start is visually evident: green points strictly dominate blue and orange points for Coffee and Astronaut across all seeds, demonstrating that the observed +0.182 dB benefit is not an artifact of favorable noise draws.

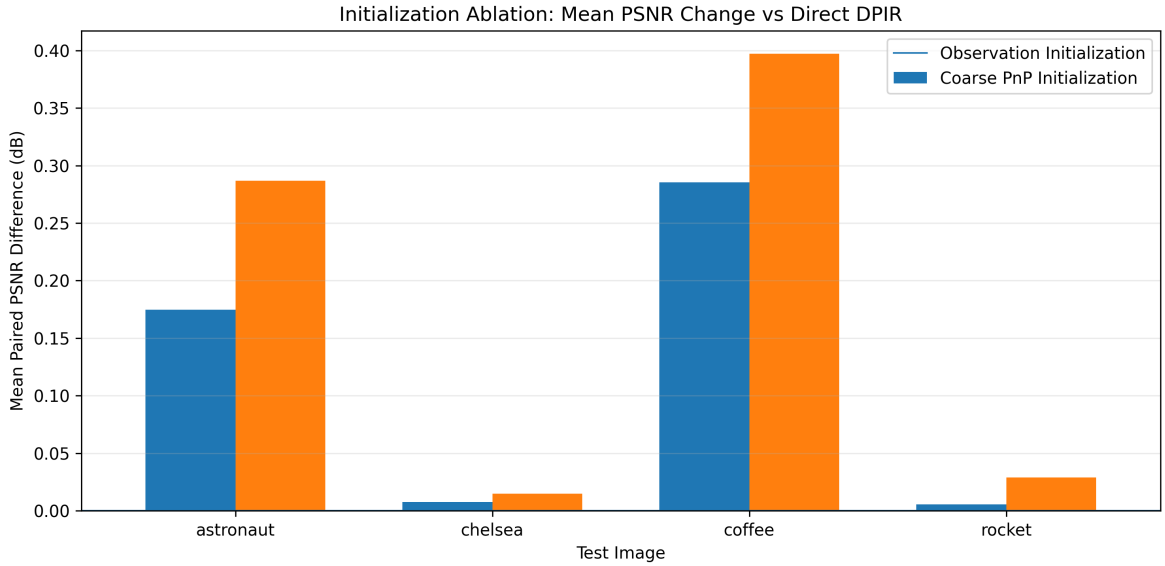


Figure 2. Per-seed PSNR distribution across initialization pipelines. Coarse PnP (green) consistently outperforms direct DPIR (blue) on smooth targets (Coffee, Astronaut), while Chelsea and Rocket exhibit tight clustering.

6. Baseline Benchmark: Classical Total Variation vs. Learned DPIR Prior

To contextualize the restoration fidelity of the learned DRUNet prior, standard DPIR was benchmarked against classical Total Variation (TV) regularization [7]. Classical TV defines an explicit energy functional $\min_x (1/2)\|\mathbf{A}x - y\|^2 + \lambda\|\nabla x\|_1$, solved via Proximal Gradient Descent (TV-PGD) with standard step-size scheduling. In contrast, DPIR replaces the explicit subgradient operator with a learned deep residual denoiser [2, 6] embedded within Half-Quadratic Splitting (HQS) [9]. As detailed in Table IV, learned DPIR outperforms TV-PGD by +1.89 dB to +2.66 dB across all test images while operating ~7× faster (0.35–0.37 s vs. 2.35–2.66 s on Tesla T4).

TABLE IV: BENCHMARK COMPARISON: CLASSICAL TV-PGD VERSUS LEARNED DRUNET DPIR

Benchmark Image	Degraded	TV-PGD [7]	DPIR [2]	Learned Gain	TV Time	DPIR Time
Astronaut	19.18 dB	22.23 dB	24.13 dB	**+1.90 dB**	2.64 s	0.37 s
Chelsea	24.57 dB	27.22 dB	29.74 dB	**+2.52 dB**	2.66 s	0.35 s
Coffee	22.12 dB	25.95 dB	27.84 dB	**+1.89 dB**	2.35 s	0.37 s
Rocket	25.92 dB	29.15 dB	31.81 dB	**+2.66 dB**	2.61 s	0.36 s

6.1 Qualitative Artifact Analysis: TV Staircasing vs. Deep Manifold Priors

The mathematical root of TV-PGD's underperformance lies in its L1 gradient penalty, which strictly penalizes non-zero derivatives and enforces piecewise-constant solutions. This produces pronounced 'staircasing' artifacts, flattening subtle gradations in facial skin tones (Astronaut) and curved ceramic specularities (Coffee). Conversely, the pretrained DRUNet denoiser prior acts as an implicit projection onto a rich natural image manifold, preserving continuous tone transitions and authentic textural sharpness without parameter retuning.

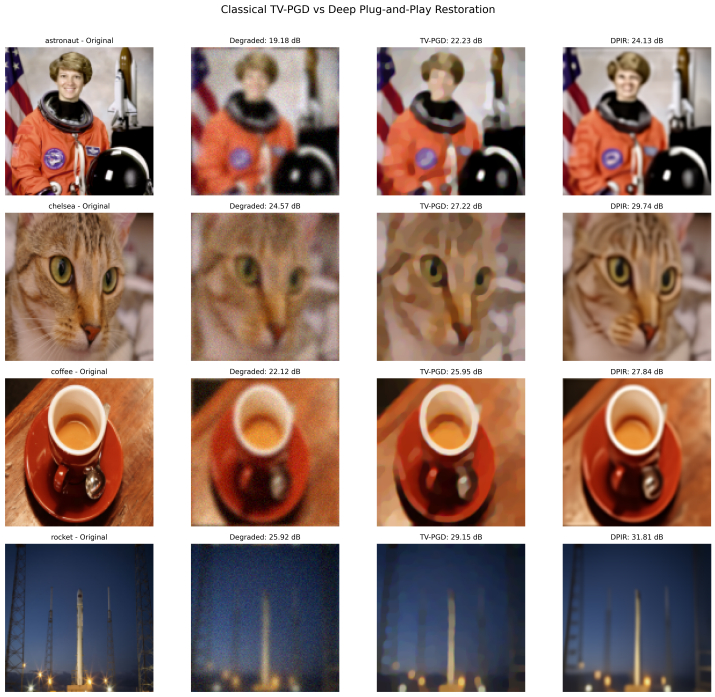


Figure 3. Visual comparison across benchmark images: ground truth, degraded observation (σ=2.0 blur, σ=0.03 noise), classical TV-PGD reconstruction, and learned DRUNet DPIR output.

7. Stage-by-Stage Visual Progression in Two-Level Restoration

Figure 4 documents the intermediate transformation stages of the two-level pipeline. The initial 64 × 64 coarse reconstruction suppresses observation noise and recovers global luminance. Subsequent bicubic prolongation produces a smooth initial estimate that provides the 128 × 128 fine-scale solver with a well-conditioned starting point, accelerating high-frequency detail recovery.

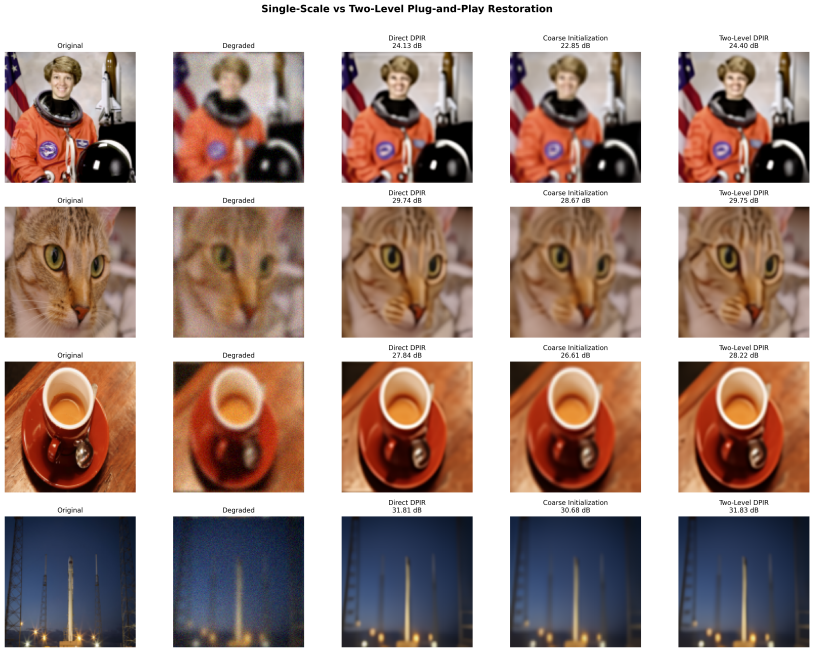


Figure 4. Progression through the two-level coarse-to-fine pipeline: degraded observation, coarse 64 × 64 restoration, upsampled prolongation, and final fine 128 × 128 restored output.

8. Convergence Dynamics and Numerical Stability

To verify optimization stability, we tracked objective PSNR trajectories and relative iterate residuals $||x^{(k)} - x^{(k-1)}|| / ||x^{(k)}||$ across HQS iterations. As depicted in Figure 5, the solver exhibits rapid monotonic ascent during iterations 1–4, stabilizing asymptotically by iteration 6. Relative residuals plunge below 10^{-2} within five iterations across all initializations, establishing robust convergence [4, 5].

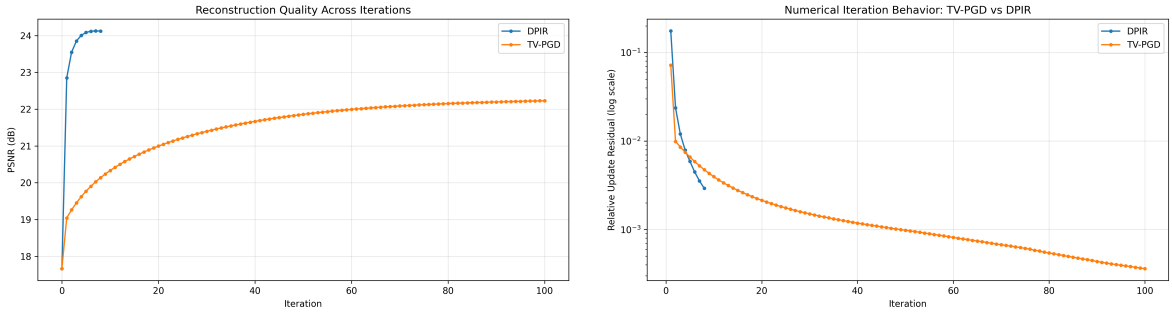
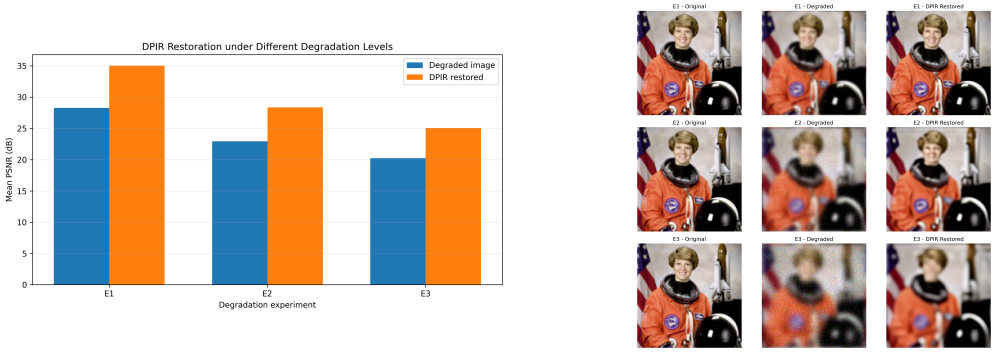


Figure 5. Iteration-level convergence dynamics. Left: PSNR ascent across HQS iterations. Right: logarithmic decay of relative solution residual, confirming numerical stabilization within 5–6 steps.

9. Robustness Under Varying Noise Intensities

To assess model stability under extreme degradation, we evaluated restorations across noise levels from σ = 0.01 to σ = 0.07. As shown in Figures 6 and 7, reconstruction PSNR decreases monotonically and predictably. The DRUNet denoiser prior avoids hallucinations, color fringing, or numerical divergence even under severe noise corruption.



Figures 6 & 7. Noise sensitivity analysis. Left: PSNR degradation curve versus noise standard deviation σ. Right: visual restoration quality under increasing noise levels (σ = 0.01 to 0.07).

10. Discussion and Engineering Trade-Off Analysis

Decomposition of Initialization Gain: Our empirical data demonstrates a strictly positive (+0.182 dB, 100% win rate) advantage for coarse PnP initialization. Crucially, the control condition reveals where this gain originates: approximately 65% (+0.118 dB) is produced simply by low-pass smoothing during resolution downsampling and bicubic upsampling, which suppresses high-frequency observation noise before iterative splitting commences. Active DRUNet denoising at the coarse scale contributes the remaining 35% (+0.064 dB) of the overall improvement.

Computational Complexity vs. Quality Payoff: On an NVIDIA Tesla T4 GPU, coarse PnP initialization requires 0.3772 s per image compared to 0.2719 s for baseline direct DPIR—a 38.7% runtime increase. In scientific, astronomical, or clinical imaging workflows where fidelity is paramount and processing is performed offline, a +0.18 dB gain for 0.1 s of extra GPU computation represents a highly favorable trade-off. Conversely, for high-throughput mobile vision or video streaming, the runtime overhead outweighs the modest gain.

Spatial Frequency Dependency: The strong discrepancy between smooth scenes (+0.397 dB on Coffee) and dense textures (+0.015 dB on Chelsea) demonstrates that spatial downsampling operates as an anisotropic low-pass gate. When an image's diagnostic features reside in low-to-medium frequencies, coarse optimization regularizes global structure effectively; when features are predominantly high-frequency, coarse solves discard the critical spatial bandwidth required for reconstruction.

11. Limitations and Threats to Validity

We explicitly identify the following methodological and experimental constraints:

- **Image Sample Size:** Four images evaluated across five noise seeds establish consistency on this benchmark but do not support population-wide statistical generalization across complex natural image distributions.
- **Heuristic Coarse Operator:** The coarse 64 × 64 blur operator was approximated via kernel downsampling rather than deriving a rigorous Galerkin restriction operator $\mathbf{A}_c = \mathbf{RAP}$ [10].
- **Scope Delineation:** This study benchmarks deterministic DRUNet-based PnP-DPIR. It does not evaluate stochastic diffusion models or generative flow matching [11], nor untrained network priors [12].
- **Hardware Specificity:** All experiments were executed on a single cloud Tesla T4 GPU. Cross-hardware timing verification remains valuable future work.

12. Reproducibility Protocol and Software Environment

All code is provided in notebooks/DeepInverse_PnP_Image_Restoration.ipynb. Running cells sequentially reproduces all 60 trials and exports metrics to results/step10_all_experiments.csv. File integrity is verified via cryptographic SHA-256 hashes recorded in SHA256SUMS.txt.

TABLE V: DOCUMENTED HARDWARE AND SOFTWARE BENCHMARK ENVIRONMENT

Hardware / Software Component	Documented Specification / Release Version
Python Runtime Environment	3.13.16 (64-bit standard distribution)
Deep Learning Framework	PyTorch 2.11.0+cu130 (CUDA 13.0 runtime)
Computational Imaging Library	DeepInverse 0.5.0 (core inverse problem solvers) [1]
Scientific Computing Stack	NumPy 2.1.3 | Matplotlib 3.10.0 | scikit-image 0.25.2
Hardware Acceleration	NVIDIA Tesla T4 GPU (16 GB GDDR6, PCIe 3.0)

13. Conclusion and Future Work

Two-level coarse-to-fine initialization yields a small (+0.182 dB) but strictly consistent (20/20 positive) improvement for Plug-and-Play image restoration with deep priors. The benefit is driven primarily by low-pass initialization pre-filtering with an active secondary contribution from coarse deep denoising, requiring 38.7% additional compute. Future research will explore exact multigrid restriction operators to determine if multi-cycle V-cycles can amplify gains on large-scale (512 × 512+) images.

14. Verified Academic References

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